

Adaptive Spectral Defense for Real-Time Recovery of Deep-Learning-Based Modulation Classifiers Under Adversarial Attack

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Abstract—Deep-learning-based automatic modulation classifiers (AMC) achieve high accuracy under benign channel conditions but collapse under adversarial perturbations crafted by optimization-based attacks such as Carlini–Wagner (CW) and Elastic-net Attack (EAD). Existing defenses require model retraining or provide insufficient accuracy recovery in the signal domain. We propose Adaptive-K v2, a unified frequency-domain defense that filters adversarial perturbations without retraining the classifier. The method combines three mechanisms in a single shared-FFT pass: spectral flatness routing for wideband modulations, SNR-adaptive K-cap to balance filtering aggressiveness with signal preservation, and per-sample magnitude-knee estimation to adapt to each modulation’s spectral width. We evaluate against five attacks (CW, EAD-L1, EAD-EN, FGSM, PGD) on RML2016.10a using the pretrained Adaptive Wavelet Network (AWN) classifier and compare against five calibrated classical filters (Kalman, Wiener, Savitzky-Golay, Gaussian, FIR) and Randomized Smoothing. Optimization attacks are devastating: CW reduces accuracy to 35.4% and EAD-L1 to 6.6%. Adaptive-K v2 recovers CW accuracy to 71.3% (+35.9 pp) and EAD-L1 to 70.8% (+64.2 pp), outperforming the best classical filter (Kalman) by 6.2 pp on CW and 8.5 pp on EAD-L1. On gradient attacks, Adaptive-K v2 recovers PGD to 43.4% (+10.9 pp) and FGSM to 52.0% (+3.9 pp). Classical filters are counterproductive on FGSM, reducing accuracy below the undefended baseline despite per-SNR calibration. These results demonstrate that a single adaptive spectral method outperforms all classical signal processing baselines across both optimization and gradient attack families. The defense adds less than 0.1 ms latency overhead per batch and degrades clean accuracy by <0.5 pp from the 92.6% baseline.

Index Terms—adversarial machine learning, automatic modulation classification, input transformation defense, spectral filtering, RF security

I. INTRODUCTION

Automatic modulation classification (AMC) is a fundamental capability in software-defined radio, cognitive radio networks, and spectrum monitoring systems [1], [2]. By identifying the modulation format of received signals without prior coordination with the transmitter, AMC enables dynamic spectrum access, interference management, and electronic surveillance. Recent deep learning approaches—including convolutional networks [3], CLDNN architectures [4], and the Adaptive Wavelet Network (AWN) [5]—achieve classification accuracy above 91% on the standard RML2016.10a benchmark across 11 modulation classes. However, the same properties

that make deep neural network (DNN) classifiers effective render them vulnerable to adversarial perturbations: small, carefully crafted modifications to IQ signals can collapse accuracy to near 0% [6]–[8].

The adversarial threat to AMC systems is qualitatively different from the image classification setting that spawned most adversarial ML research [9], [10]. RF IQ signals have amplitudes of approximately ± 0.02 , so even small L_∞ perturbations can constitute a large fraction of the signal dynamic range. Optimization-based attacks such as the Carlini–Wagner L_2 attack (CW) [11] and the Elastic-Net Attack on Deep Neural Networks (EAD) [12] are particularly effective: they reduce AWN accuracy from above 90% to effectively 0% at SNR 18 dB. Existing defenses for AMC fall into two categories: adversarial training, which requires retraining the classifier on attack-generated examples and must be repeated for each new attack [13], [14]; and input-transformation defenses, which preprocess signals before classification and require no model modification [15], [16]. However, no systematic comparison has been made between frequency-domain adaptive methods and the classical signal processing filters—Kalman, Wiener, Savitzky-Golay, Gaussian, and FIR—that communications engineers would naturally apply as baselines [17], [18].

We propose an adaptive spectral defense pipeline that operates entirely in the frequency domain. Our key insight is that adversarial perturbations on IQ signals tend to occupy the high-frequency spectral components of the signal rather than the low-frequency components that carry modulation information. By applying a fast Fourier transform (FFT) and retaining only the K largest magnitude bins, we filter out adversarial energy while preserving the underlying modulation signature. We augment this with a spectral flatness routing criterion that detects wideband modulations (AM-SSB) where Top- K filtering would destroy signal content, and an adaptive- K mechanism that automatically selects per-sample filtering intensity based on the spectral magnitude knee-point. We benchmark all proposed methods against five calibrated classical signal processing filters and randomized smoothing [19], providing the first systematic evaluation of this design space for AMC defense.

This paper makes four contributions:

- 1) **Control-plane attack characterization:** We experimentally confirm that adversarial attacks on AMC are *control-plane attacks*—they cause the classifier to select

the wrong demodulator without corrupting the underlying signal. Across five attack algorithms (CW, EADL1, EADEN, FGSM, PGD) and eight digital modulations, CRC integrity remains above 71% under oracle demodulation at SNR 18 dB, demonstrating that the adversarial perturbation leaves the data waveform intact. This motivates frequency-domain defense: if the perturbation is a spectral artifact, filtering can remove it.

- 2) **Adaptive- K FFT defense:** We propose automatic per-sample spectral filtering intensity selection via a magnitude knee-point detection algorithm. Rather than using a fixed K for all signals, the adaptive- K method sorts FFT magnitudes in descending order and finds the elbow where cumulative energy exceeds a threshold, setting K equal to the corresponding index. This adapts filtering to signal bandwidth, preserving narrow-band modulations (BPSK: small K) while allowing wider-band signals the frequency support they require.
- 3) **Systematic classical filter comparison:** We provide the first evaluation of five calibrated classical signal processing filters—Kalman, Wiener, Savitzky-Golay, Gaussian, and FIR—as adversarial defenses for AMC. Filter parameters are calibrated per-SNR via grid search on validation accuracy. Our adaptive spectral methods outperform all classical baselines by 4–8% on optimization-based attacks (CW, EAD) at high SNR, while classical filters show competitive performance at low SNR where adversarial attacks are less effective than channel noise.
- 4) **Unified adaptive spectral defense (Adaptive- K v2):** We combine three mechanisms in a single shared-FFT pass: (i) spectral flatness routing for wideband modulations (AM-SSB) to input quantization, (ii) SNR-adaptive K -cap that adjusts filtering aggressiveness based on estimated signal quality, and (iii) per-sample magnitude-knee estimation that adapts to each modulation’s spectral width. This single method outperforms all classical baselines on four of five attacks, recovering CW accuracy from 35.4% to 71.3% (+35.9 pp) and EAD-L1 from 6.6% to 70.8% (+64.2 pp).

The remainder of this paper is organized as follows. Section II reviews related work on adversarial AMC attacks, defense methods, and classical signal processing filters. Section III presents the system model, attack formulations, and threat model. Section IV describes the proposed Adaptive- K v2 defense and classical filter baselines. Section V details the experimental setup. Section VI presents results and analysis. Section VII concludes the paper.

II. RELATED WORK

A. Adversarial Attacks on AMC Systems

The vulnerability of deep learning AMC classifiers to adversarial perturbations was first systematically examined by Lin et al. [8], who applied FGSM and PGD to DNN-based modulation recognition, demonstrating that single-step and iterative gradient attacks can degrade accuracy from above 90% to below 10%. Flowers et al. [20] provided a comprehensive evaluation of adversarial evasion attacks in the wireless

context, comparing multiple attack strengths and showing that the threat is practical under realistic signal conditions. Yang et al. [21] characterized the geometry of adversarial examples for wireless DNN classifiers, finding that the decision boundary is significantly closer to clean samples in the IQ domain than in equivalent image classification tasks.

Bahramali et al. [22] introduced over-the-air adversarial attacks that survive the physical channel, revealing a more stringent threat model where perturbations must remain effective after transmission, fading, and noise. Silvaco et al. [7] provided a comprehensive survey of adversarial attacks against AMC, categorizing them by perturbation norm (L_∞ , L_2 , L_1), attack objective (untargeted vs. targeted), and knowledge assumption (white-box vs. black-box). Zhang et al. [23] introduced stealthy adversarial attacks that minimize spectral anomaly while maintaining high misclassification rates, posing challenges for energy-based detectors. Kim et al. [24] explored frequency-domain adversarial attacks that concentrate perturbation energy in specific spectral bands, making conventional band-limited filtering less effective.

Among attack methods, optimization-based attacks (CW [11], EAD [12]) consistently outperform gradient-based approaches (FGSM [9], PGD [10]) on IQ signals, achieving near-zero accuracy against the AWN classifier at high SNR with minimal perturbation norm. This is because IQ signals have low signal amplitude (~ 0.02) relative to their dynamic range, allowing optimization-based methods to find highly effective perturbations near the classifier decision boundary. In contrast to prior work that focuses solely on attacks, we use these attacks as benchmarks to evaluate defense recovery.

B. Defense Methods for AMC

Defense strategies for AMC can be broadly divided into model-level and input-level approaches. *Model-level* defenses modify the classifier through adversarial training [13], incorporating adversarial examples during training to improve robustness. Tramèr et al. [14] extended this with ensemble adversarial training using transfers from multiple surrogate models. These methods are effective against specific attacks but require retraining for each new threat, limiting deployability in operational systems where the classifier is a fixed checkpoint. Papernot et al. [25] proposed defensive distillation to smooth classifier gradients, reducing attack effectiveness, but this approach has been shown to be bypassable by adaptive attacks.

Input-level (input-transformation) defenses preprocess the input before classification and require no model modification. Guo et al. [15] systematically evaluated input transformations for image adversarial defense, including JPEG compression, total variation minimization, and image quilting, finding that FFT-based filtering is among the most effective. Xu et al. [16] proposed feature squeezing, which reduces the feature space via color depth reduction and spatial smoothing, then uses disagreement between original and squeezed predictions as a detection signal. Dziugaite et al. [26] studied JPEG compression as a defense, showing that lossy compression removes adversarial high-frequency components while preserving seman-

tic content. These image-domain insights motivate frequency-domain filtering for IQ signals.

Cohen et al. [19] introduced randomized smoothing for certified robustness: by averaging classifier predictions over Gaussian-noised copies of the input, the classifier obtains a certified L_2 robustness guarantee. While randomized smoothing provides formal robustness certificates, its computational cost (many forward passes per sample) makes real-time deployment challenging. We include randomized smoothing as a certified baseline for comparison with our deterministic filtering approach.

Recent work has applied frequency-domain ideas specifically to RF defense. Kim et al. [24] proposed band-selective suppression based on detecting spectral anomalies introduced by adversarial perturbations. Our adaptive- K method differs by operating on the full spectrum without prior knowledge of which band the attack targets, making it attack-agnostic. No prior work has performed a systematic side-by-side comparison of frequency-domain adaptive methods and classical communications signal processing filters as adversarial defenses.

C. Classical Signal Processing Filters

Classical signal processing offers a rich toolkit of denoising filters developed for communications and estimation problems. The Wiener filter [27] is the minimum mean square error (MMSE) estimator under the assumption of stationary signal and additive noise, operating in the frequency domain by applying a magnitude-squared SNR weighting to each spectral component. Kalman filtering [28] extends Wiener estimation to non-stationary signals via a state-space model, recursively tracking signal evolution with prediction and update steps. Both filters are well-suited to thermal noise but were not designed for adversarial perturbations, which are structured rather than random.

Savitzky-Golay smoothing [29] fits a polynomial of specified degree to a sliding window of samples using local least-squares, producing a smoothing filter that preserves the shape of peaks and pulse signals. Unlike frequency-domain filters, Savitzky-Golay operates in the time domain and is particularly effective at preserving high-frequency signal features while suppressing narrow noise spikes. Gaussian and FIR lowpass filters are standard tools in digital communications for pulse shaping and anti-aliasing; their regularization effect on IQ signals has been studied in the context of matched filtering [30] and channel estimation [17], though their use as adversarial defenses has not been examined in the AMC context.

The application of classical filters as adversarial defenses rests on the hypothesis that adversarial perturbations behave like structured noise that can be attenuated by denoising. However, this hypothesis is non-trivial: unlike AWGN, adversarial perturbations are deliberately crafted to align with the classifier gradient and may therefore concentrate energy in exactly the spectral regions where the modulation signal is strongest. Our experiments test this hypothesis directly by evaluating calibrated classical filters against optimization-based attacks (CW, EAD) on the RML2016.10a dataset.

In contrast to the above works, we propose an adaptive spectral defense that combines FFT Top- K filtering with

per-sample spectral flatness routing, and provide the first systematic comparison of frequency-domain adaptive methods against five calibrated classical filters and randomized smoothing as adversarial defenses for AMC. Our approach achieves the best accuracy-latency trade-off among all evaluated methods while remaining model-agnostic and suitable for real-time deployment.

III. SYSTEM MODEL AND THREAT MODEL

A. Signal Representation and Dataset

We consider complex baseband IQ signals represented as real-valued tensors $\mathbf{x} \in \mathbb{R}^{2 \times T}$, where the first dimension indexes the in-phase (I) and quadrature (Q) channels and T is the number of time samples. Each channel captures a different aspect of the baseband signal; their joint representation encodes the amplitude and phase of the modulated carrier. Following the RML2016.10a dataset convention [3], we use $T = 128$ samples, corresponding to 16 symbols at 8 samples-per-symbol with a root-raised-cosine pulse shape.

The RML2016.10a dataset contains 220,000 IQ bursts drawn from 11 modulation classes: BPSK, QPSK, 8PSK, QAM16, QAM64, PAM4, AM-DSB, AM-SSB, WBFM, GFSK, and CPFSK. Each burst is labeled with the modulation type and the signal-to-noise ratio (SNR) at which it was recorded, spanning -20 to $+18$ dB in 2 dB steps. At high SNR (≥ 0 dB), the AWN classifier [5] achieves above 91% overall accuracy across all 11 classes.

B. AWN Classifier Architecture

The Adaptive Wavelet Network (AWN) [5] takes batched IQ tensors of shape $[N, 2, T]$ and predicts one of $C = 11$ modulation classes. The forward pass has three stages:

Convolutional feature extraction. A 2D convolution with kernel size $[2, 7]$ integrates the I and Q channels, yielding a feature map of shape $[N, 64, 1, T]$, which is then squeezed to $[N, 64, T]$. A second 1D convolution with batch normalization [31] and LeakyReLU extracts temporal features, producing $[N, 64, T]$ feature maps.

Adaptive wavelet decomposition. The core module applies a learnable lifting scheme [5] that recursively splits the feature map into even and odd subsequences, applies learnable Predictor (P) and Updater (U) operators to produce approximation coefficients $\mathbf{c}$ and detail coefficients $\mathbf{d}$, and repeats for `num_levels` levels. The detail regularization loss is:

$$\mathcal{R}_d = \sum_{\ell=1}^L \lambda_d \|\mathbf{d}_\ell\|_2^2, \quad \mathcal{R}_c = \lambda_c |\bar{\mathbf{c}}|, \quad (1)$$

where $\bar{\mathbf{c}}$ is the mean of the final approximation. The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{R}_d + \mathcal{R}_c, \quad (2)$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy classification loss.

Attention and classification. Squeeze-excitation (SE) attention [32] computes per-level weights for the multi-scale

approximation and detail coefficient tensors. The attention-weighted features are pooled and passed through two fully-connected layers with dropout and LeakyReLU activations to produce the final class logits. The model is trained using the Adam optimizer [33] with early stopping.

C. Attack Models

We evaluate four attacks that span the gradient-based and optimization-based attack families. All attacks use `torchattacks` [34] with per-sample minmax normalization: each IQ sample is mapped to $[0, 1]$ using $\tilde{x} = (x - x_{\min}) / (x_{\max} - x_{\min})$ before attack generation and mapped back afterward. This makes epsilon values interpretable relative to the signal range.

FGSM. The Fast Gradient Sign Method [9] computes a single gradient step:

$$\mathbf{x}_{\text{adv}} = \mathbf{x} + \varepsilon \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}_{\text{CE}}(\mathbf{x}, y)), \quad (3)$$

where ε bounds the L_{∞} perturbation magnitude. FGSM is computationally efficient but produces suboptimal adversarial examples.

PGD. Projected Gradient Descent [10] iterates multiple gradient steps with projection onto the ε -ball:

$$\mathbf{x}^{(t+1)} = \Pi_{B_{\varepsilon}(\mathbf{x})}(\mathbf{x}^{(t)} + \alpha \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}_{\text{CE}}(\mathbf{x}^{(t)}, y))), \quad (4)$$

where Π denotes projection and α is the step size. PGD with a random start is regarded as a strong first-order attack [35].

CW (L_2). The Carlini–Wagner attack [11] solves the constrained optimization:

$$\min_{\delta} \|\delta\|_2 + c f(\mathbf{x} + \delta), \quad (5)$$

where $f(\cdot)$ is a surrogate misclassification objective and c balances perturbation norm against attack success. CW distributes adversarial energy broadly across spectral frequencies, making it resistant to simple band-limited filtering. We use $c = 10, 200$ optimization steps.

EAD (L_1 and EN). The Elastic-Net Attack on Deep Neural Networks [12] adds an L_1 regularization term to the CW objective:

$$\min_{\delta} \beta \|\delta\|_1 + \|\delta\|_2 + c f(\mathbf{x} + \delta), \quad (6)$$

where β controls the sparsity of the perturbation. We evaluate both the L_1 variant (EADL1, β large) and the elastic-net variant (EADEN, β moderate). EADL1 produces sparse perturbations concentrated in few spectral components, while EADEN produces a mixture of sparse and distributed perturbations.

D. Threat Model

Attacker capabilities. We consider a white-box adversary with full knowledge of the AWN classifier architecture, trained weights, and input normalization procedure. The attacker can query the classifier and compute exact gradients. This is the strongest adversary assumption and establishes a worst-case defense baseline.

Attacker goal. The adversary seeks untargeted misclassification: any prediction other than the true modulation class is

considered a success. The attacker does not attempt to cause the classifier to predict any specific wrong class.

Attacker constraint. Perturbations are bounded by the chosen attack norm: $\|\delta\|_{\infty} \leq \varepsilon$ for FGSM/PGD, $\|\delta\|_2 \leq$ (implicitly, via CW minimization) for CW, and elastic-net bounded for EAD. We evaluate $\varepsilon \in \{0.05, 0.1\}$ in the per-sample minmax-normalized space (equivalent to 5–10% of the per-sample signal range).

Defense-unaware model. We assume the attacker does not adapt to the defense mechanism: adversarial examples are generated against the undefended AWN classifier. This models the realistic deployment scenario where the defense is applied transparently at the receiver and is not publicly disclosed. Adaptive attacks against the defense are left for future work.

Defender capabilities. The defender can preprocess each IQ burst before classification, with access only to the received signal. The defender does not know the true modulation class, the attack type, or the attack parameters. The defense must operate in real time: latency must be sub-millisecond per sample to meet the requirements of practical AMC deployment contexts, including ITU/FCC spectrum monitoring stations, CBRs Environmental Sensing Capability (ESC) nodes, and military electronic support measures (ESM) systems [30]. In each of these contexts, AMC drives downstream decisions (spectrum allocation, threat identification) that require near-real-time throughput, typically at 25–100 Msps IQ digitization rates with burst processing latency budgets of 1–10 ms per channel.

IV. PROPOSED DEFENSE METHODS

A. Control-Plane Attack Insight

Before describing our defenses, we motivate the frequency-domain approach by establishing the nature of adversarial attacks on AMC. We conduct a two-track experiment to separate *classification failure* from *waveform corruption*. In Track A, we apply CW and EAD attacks to real RML2016.10a signals and record classification accuracy. In Track B, we generate synthetic bursts with known payload bits and CRC-8 checksums using a full TX/RX chain (pulse shaping, AWGN channel, matched filtering, pilot-based equalization), transfer the adversarial perturbation vectors from Track A, and evaluate CRC pass rate under four demodulation conditions:

- 1) Clean+Oracle: clean signal with true modulation known;
- 2) Clean+AMC: clean signal, modulation predicted by AWN;
- 3) Adv+Oracle: adversarial signal with true modulation known;
- 4) Adv+AMC: adversarial signal, modulation predicted by AWN.

Table I shows CRC pass rates at SNR 18 dB under CW attack for eight digital modulations. The Adv+Oracle column reveals that the underlying signal remains largely intact under adversarial perturbation: CRC pass rates stay above 71% when the correct demodulator is applied. The catastrophic failure manifests only in the Adv+AMC column (0–39%), where the classifier selects the wrong demodulator.

TABLE I: CRC Pass Rate (%) at SNR 18 dB — CW Attack. Adversarial perturbation drops AMC accuracy to near 0% (Adv+AMC) but CRC integrity under oracle demodulation (Adv+Oracle) remains above 71%, confirming that attacks target the classification decision, not the physical signal.

Mod	Clean+Orc	Clean+AMC	Adv+Orc	Adv+AMC
BPSK	100.0	97.0	100.0	39.0
QPSK	100.0	98.5	100.0	1.5
8PSK	100.0	98.0	100.0	0.5
QAM16	100.0	94.5	99.5	0.5
QAM64	96.0	94.0	76.0	0.5
PAM4	100.0	98.5	71.0	2.0
CPFSK	100.0	100.0	100.0	18.5
GFSK	99.5	99.0	99.5	0.0

This *control-plane attack* insight motivates our frequency-domain approach: adversarial perturbations are spectral artifacts that can in principle be attenuated by frequency-domain filtering without destroying the underlying modulation waveform.

B. Adaptive-K v2: Unified Spectral Defense

We propose Adaptive-K v2, a unified defense that combines three mechanisms in a single shared-FFT pass: (1) wideband routing via spectral flatness, (2) SNR-adaptive K-cap, and (3) per-sample magnitude-knee estimation. The method operates directly on raw IQ data without normalization, since FFT Top- K selects bins by relative magnitude (scale/shift invariant).

1) *FFT Top- K Filtering*: The fundamental building block is FFT Top- K filtering, which retains the K largest-magnitude spectral components and zeros all others:

$$\hat{\mathbf{x}} = \text{IFFT}(\text{TopK}(\text{FFT}(\mathbf{x}), K)). \quad (7)$$

Adversarial perturbations distribute energy across the full spectrum; retaining only the dominant K components removes this diffuse adversarial energy while preserving the modulation-carrying spectral components.

2) *Wideband Routing via Spectral Flatness*: FFT Top- K is effective for narrowband modulations (BPSK, QPSK, QAM, etc.) but fails for wideband modulations such as AM-SSB, whose spectral energy is spread uniformly. We exploit *spectral flatness* to detect and route wideband signals:

$$\text{SF}(\mathbf{x}) = \frac{\exp(\frac{1}{N} \sum_k \ln |X_k|^2)}{\frac{1}{N} \sum_k |X_k|^2}. \quad (8)$$

Narrowband modulations have $\text{SF} < 0.04$; AM-SSB has $\text{SF} \approx 0.55$ —a gap of more than $18\times$ providing robust routing. Samples with $\text{SF} > 0.4$ are routed to per-sample quantization ($L = 32$ levels) instead of Top- K .

3) *SNR-Adaptive K-Cap*: Per-sample K is estimated via the magnitude knee: the smallest index where the sorted spectral magnitude drops below 5% of the peak. However, two problems arise: (1) adversarial perturbations inflate the spectrum, pushing the knee to larger K values that retain adversarial bins; (2) at low SNR, aggressive filtering (small K) removes signal along with noise.

Algorithm 1 Adaptive-K v2: Unified Spectral Defense

Require: Input $\mathbf{x} \in \mathbb{R}^{N \times 2 \times T}$, flatness threshold $\tau = 0.4$, quantization levels $L = 32$, knee ratio $\rho = 0.05$, pilot bins $K_p = 10$

Ensure: Defended signal $\hat{\mathbf{x}}$

```

1:  $\mathbf{X} \leftarrow \text{FFT}(\mathbf{x})$  // ONE shared FFT
2: for each sample  $i = 1, \dots, N$  do
3:    $\text{SF}_i \leftarrow \text{SpectralFlatness}(\mathbf{X}_i)$  // Eq. (8)
4:   if  $\text{SF}_i > \tau$  then // wideband (AM-SSB)
5:      $\hat{\mathbf{x}}_i \leftarrow \text{Quantize}(\mathbf{x}_i, L)$ 
6:   else // narrowband
7:      $\widehat{\text{SNR}}_i \leftarrow \text{Eq. (9)}$  // from same FFT
8:      $K_{\max, i} \leftarrow \text{Eq. (10)}$  // SNR-adaptive cap
9:      $\text{knee}_i \leftarrow \text{first } k \text{ where } |\mathbf{X}_i^\downarrow[k]|/|\mathbf{X}_i^\downarrow[1]| < \rho$ 
10:     $K_i \leftarrow \min(\text{knee}_i, K_{\max, i})$ 
11:     $\hat{\mathbf{x}}_i \leftarrow \text{IFFT}(\text{TopK}(\mathbf{X}_i, K_i))$  // reuse FFT
12:   end if
13: end for
14: return  $\hat{\mathbf{x}}$ 
    
```

We address both by estimating SNR from the same FFT and mapping it to a per-sample K cap:

$$\widehat{\text{SNR}}_i = \frac{\sum_{k=1}^{K_p} P_i^\downarrow[k]}{\sum_{k=K_p+1}^T P_i^\downarrow[k]}, \quad (9)$$

where $P_i^\downarrow$ is the sorted power spectrum and $K_p = 10$ bins serve as signal proxy. The cap is then:

$$K_{\max, i} = \begin{cases} 35 & \text{if } \widehat{\text{SNR}}_i \leq 3 \text{ (low SNR, gentle)} \\ 20 & \text{if } \widehat{\text{SNR}}_i \geq 10 \text{ (high SNR, aggressive)} \\ \text{linear interp.} & \text{otherwise} \end{cases} \quad (10)$$

The final K per sample is $K_i = \min(\text{knee}_i, K_{\max, i})$.

4) *Complete Algorithm*: Algorithm 1 presents the unified defense. All steps share a single FFT computation.

The total cost is one FFT plus one IFFT, or $\mathcal{O}(T \log T)$ per sample for $T = 128$. Each mechanism addresses a specific failure mode: wideband routing handles AM-SSB (where Top- K destroys the signal), the SNR-adaptive cap prevents adversarial bin inflation at high SNR while preserving weak signals at low SNR, and the per-sample knee adapts to each modulation's natural spectral width (GFSK uses $K \approx 17$, QPSK uses $K \approx 25$).

C. Classical Filter Baselines

To contextualize the performance of our frequency-domain adaptive methods, we evaluate five classical signal processing filters as adversarial defense baselines. These represent the standard signal processing toolkit that a communications engineer would apply before considering ML-specific defenses. All filters are calibrated per-SNR via grid search over their hyperparameters on a held-out validation subset of RML2016.10a, maximizing post-filter classification accuracy on the undefended model.

Kalman filter. The Kalman filter [28] models each IQ channel as a first-order random walk corrupted by observation

noise, with process noise covariance Q and observation noise covariance R as tunable parameters. The scalar filter recursively estimates the signal state using prediction and update steps. For AMC defense, R is calibrated to the noise floor of adversarial perturbations. Kalman filtering is optimal for linear Gaussian noise but was not designed for structured adversarial perturbations; our experiments test whether its denoising properties generalize.

Wiener filter. The Wiener filter [27] computes the MMSE estimate in the frequency domain by applying a signal-to-noise weighting $H_k = S_{ss,k}/(S_{ss,k} + S_{nn,k})$ to each spectral bin, where $S_{ss,k}$ and $S_{nn,k}$ are the signal and noise power spectral densities at frequency k , estimated from calibration data. The Wiener filter assumes stationary signal and noise statistics; adversarial perturbations violate the stationarity assumption.

Savitzky-Golay filter. The Savitzky-Golay filter [29] smooths each IQ channel by fitting a polynomial of degree d to a sliding window of w samples using local least squares. The filter preserves the shape of peaks and pulses better than standard lowpass filters, making it suitable for preserving QAM constellation transitions. Parameters d and w are calibrated via grid search.

Gaussian filter. A Gaussian kernel $h[n] = \exp(-n^2/2\sigma^2)$ is convolved with each IQ channel, implementing a smooth lowpass response with cutoff determined by the standard deviation σ . The filter is implemented as a GPU-native `F.conv1d` operation for efficient batch processing, avoiding CPU roundtrip latency.

FIR lowpass filter. A finite impulse response (FIR) lowpass filter with normalized cutoff frequency f_c and filter order M is applied to each IQ channel [18]. The filter coefficients are computed via windowed-sinc design (Hamming window). Like the Gaussian filter, the FIR filter is implemented as a depthwise `F.conv1d` convolution for GPU efficiency.

We also evaluate **randomized smoothing** [19] as a certified robustness baseline. Randomized smoothing adds isotropic Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ to the input and takes the majority vote over k noisy copies. The certified radius is $\sigma \Phi^{-1}(p_A)$, where p_A is the probability of the top class under smoothing. We use $\sigma = 0.1$ and $k = 100$ copies, which is the lowest latency configuration that provides non-trivial certificates. All five classical filters and randomized smoothing are calibrated and evaluated under identical experimental conditions.

D. Computational Complexity

The computational cost of each defense is dominated by its core operation. FFT Top- K (and adaptive- K) requires $\mathcal{O}(T \log T)$ for the real FFT, $\mathcal{O}(T)$ for magnitude computation, $\mathcal{O}(T \log K)$ for top- K selection via partial sort, and $\mathcal{O}(T \log T)$ for the inverse FFT. The spectral flatness computation adds $\mathcal{O}(T)$. Total per-sample cost is $\mathcal{O}(T \log T)$, dominated by the FFT.

Among the classical filters, Gaussian and FIR convolutions cost $\mathcal{O}(T \cdot M)$ per channel, where M is the filter length (typically 17–33 taps); both are implemented as GPU-native depthwise convolutions with negligible latency. Savitzky-Golay uses a precomputed coefficient vector and has the

same $\mathcal{O}(T \cdot w)$ complexity as a FIR filter. Kalman filtering requires a sequential state update loop, with $\mathcal{O}(T)$ operations per sample but poor parallelism (the state update at time t depends on time $t - 1$), necessitating a CPU fallback and incurring higher per-batch latency despite lower asymptotic complexity. Wiener filtering applies per-bin frequency-domain weighting in $\mathcal{O}(T \log T + T)$ using a precomputed gain vector. Randomized smoothing requires k forward passes through the classifier ($k = 100$ by default), making it $100\times$ more expensive than a single inference. Detailed latency measurements and throughput comparisons are reported in Section VI.

V. EXPERIMENTAL SETUP

A. Dataset and Model

We evaluate on the RadioML 2016.10a dataset [3], a widely used benchmark for automatic modulation classification comprising 220,000 IQ signal frames spanning 11 modulation classes: eight digital (BPSK, QPSK, 8PSK, QAM16, QAM64, PAM4, CPFSK, GFSK) and three analog (AM-DSB, AM-SSB, WBFM). Each frame consists of 128 complex-valued samples represented as a 2×128 real tensor of in-phase (I) and quadrature (Q) components. The dataset covers SNR values from -20 dB to $+18$ dB in 2 dB increments, yielding 20 SNR levels and 1,000 frames per (modulation, SNR) cell.

We partition the data using stratified sampling that preserves the joint (modulation, SNR) distribution across splits: 70% training, 15% validation, and 15% test. All reported accuracies are computed on the held-out test split.

The classifier under attack is the Adaptive Wavelet Network (AWN) [5], a pretrained model that achieves 92.6% classification accuracy at $\text{SNR} \geq 0$ dB on the clean test set. We use the publicly available pretrained checkpoint without any adversarial fine-tuning, so the model represents a realistic deployment target that has not been hardened against adversarial inputs.

B. Attack Configurations

We evaluate five attacks drawn from two families: single-step gradient attacks (FGSM), iterative gradient attacks (PGD), and optimization-based attacks (CW-L2, EAD-L1, EAD-EN). All attacks are implemented via the `torchattacks` library [34] using a `Model01Wrapper` that maps IQ signals into the $[0, 1]$ range expected by `torchattacks` and returns logits only. We use per-sample min-max normalization (`ta_box=minmax`) to map each sample to $[0, 1]$ so that epsilon is expressed relative to the sample's own dynamic range—a natural choice for IQ signals whose peak-to-peak amplitude varies widely across modulations and SNR levels.

Table II summarizes the attack hyperparameters. For FGSM and PGD, $\varepsilon = 0.03$ after min-max normalization corresponds to roughly 60% of the average per-sample dynamic range at high SNR—a moderate threat that leaves the signal recognizable but is effective enough to reduce accuracy substantially. For CW and EAD, the optimization constant c controls the trade-off between perturbation minimization and attack success; we use $c = 1.0$ and $c_0 = 0.01$, respectively, as these yield near-zero undefended accuracy on most modulations at high SNR while keeping perturbation norms in a physically plausible range.

TABLE II: Attack configurations. All attacks target untargeted misclassification in a white-box setting. Epsilon values are expressed after per-sample min-max normalization to $[0, 1]$.

Attack	Norm	Key Parameters	Source
FGSM	ℓ_∞	$\varepsilon = 0.03$	[9]
PGD	ℓ_∞	$\varepsilon = 0.03$, steps = 10, $\alpha = 0.01$	[10]
CW	ℓ_2	$c = 1.0$, steps = 100, lr = 0.01	[11]
EAD-L1	ℓ_1	$c_0 = 0.01$, $\beta = 0.01$, steps = 100	[12]
EAD-EN	elastic net	$c_0 = 0.01$, $\beta = 0.01$, steps = 100	[12]

TABLE III: Defense configurations. Classical filter parameters are calibrated per-SNR via grid search on the validation set; representative values at SNR = 10 dB are shown. Adaptive-K v2 requires no per-SNR calibration.

Defense	Type	Key Parameters (SNR = 10 dB)
Adaptive-K v2	Proposed	Per-sample K via magnitude knee; SNR-adaptive cap $K_{\max} \in [20, 35]$; wideband routing at flatness > 0.4
Kalman	Classical	$Q = 0.005$, $R = 0.001$; CPU sequential
Wiener	Classical	noise = 0.01, filter len = 3; freq. domain
Savitzky-Golay	Classical	window = 5, order = 1; time domain
Gaussian	Classical	$\sigma = 0.5$; GPU depthwise conv1d
FIR	Classical	cutoff = 0.15, taps = 63; GPU depthwise conv1d
Rand. Smoothing	Baseline	$\sigma = 0.1$, $k = 1$ copies; certified robustness

C. Defense Parameters

We evaluate eight defense configurations: the proposed Adaptive-K v2 and seven baselines (Kalman, Wiener, Savitzky-Golay, Gaussian, FIR, and Randomized Smoothing), plus the undefended baseline. The proposed method operates entirely in the frequency domain without retraining and requires no per-SNR calibration.

Table III summarizes the defense parameters. For the five classical filters (Kalman, Wiener, Savitzky-Golay, Gaussian, FIR), we perform a grid search over the parameter space at each SNR level using the validation set with CW attack. The best-performing parameter combination per SNR is stored in a calibration file (`calibration_params.json`) and loaded at test time, giving the classical baselines every possible advantage. Randomized Smoothing [19] adds Gaussian noise ($\sigma = 0.1$) before classification using a single forward pass ($k = 1$), consistent with its use as an inference-time defense rather than a certified smoothed classifier.

D. Evaluation Metrics

We report the following metrics:

Per-SNR accuracy: fraction of test samples correctly classified at each SNR level $s \in \{0, 2, \dots, 18\}$ dB. We focus on non-negative SNR because at SNR < 0 dB the clean accuracy is below 50% and adversarial perturbations are masked by channel noise.

Weighted average accuracy: the mean per-SNR accuracy, weighted by the number of test samples at each SNR level. Since RML2016.10a is balanced, this reduces to the simple mean over the 10 SNR levels ≥ 0 dB evaluated.

Perturbation budget curves: accuracy as a function of attack strength (ε for FGSM/PGD; optimization constant c for CW/EAD) to assess defense robustness across the threat spectrum.

Confusion matrices: per-class classification accuracy before and after defense at a fixed SNR (18 dB) to identify which modulation classes benefit most from the proposed methods.

All evaluations use 200 samples per (SNR, modulation) cell from the test split, yielding 2,200 samples per SNR level (200 samples $\times$ 11 modulation classes).

VI. RESULTS AND ANALYSIS

A. Defense Comparison Overview

Table IV presents the primary result: weighted-average classification accuracy for nine defense configurations against all five attacks at SNR ≥ 0 dB. The proposed Adaptive-K v2 achieves the highest accuracy on four of five attacks, with particularly large margins on the optimization-based attacks CW, EAD-L1, and EAD-EN.

Figure 1 provides a visual overview of Table IV as a grouped bar chart. Four key findings emerge:

Finding 1 — Optimization attacks are devastating without defense. CW reduces accuracy from 92.6% to 35.4%, EAD-L1 to 6.6%, and EAD-EN to 3.4%—near chance level for 11 classes. These norm-minimizing attacks produce perturbations that are imperceptible in the time domain yet catastrophically effective.

Finding 2 — Adaptive-K v2 achieves the largest recovery. On CW, Adaptive-K v2 recovers accuracy from 35.4% to 71.3% (+35.9 pp). On EAD-L1 the recovery is even larger: from 6.6% to 70.8% (+64.2 pp). The margin over Kalman (best classical filter) is 6.2 pp on CW (71.3% vs. 65.2%) and 8.5 pp on EAD-L1 (70.8% vs. 62.3%). This demonstrates that the unified adaptive spectral approach far outperforms calibrated classical filters.

Finding 3 — Classical filters are counterproductive on FGSM. On FGSM, the undefended baseline (48.1%) exceeds every classical filter: Kalman (46.8%), Gaussian (46.0%), FIR (46.4%), Wiener (45.6%), Savitzky-Golay (45.6%). This counterintuitive result occurs because FGSM perturbations are spectrally diffuse—resembling the signal itself—so time-domain smoothing removes signal energy alongside perturbation energy. Adaptive-K v2 achieves 52.0%, recovering 3.9 pp above the baseline.

Finding 4 — Randomized smoothing is unsuitable for AMC. Adding Gaussian noise ($\sigma = 0.01$) collapses accuracy to $\sim 30\%$ across all attacks. The additive noise destroys the fine-grained IQ structure that AMC classifiers rely on.

B. Accuracy vs. SNR for Optimization Attacks

Figure 2 shows per-SNR accuracy for CW and EAD-L1. Three behavioral regimes are visible:

TABLE IV: Weighted-average classification accuracy (%) across $\text{SNR} \in \{0, 2, \dots, 18\}$ dB for nine defenses against five attacks on RML2016.10a. **Bold** indicates best result per attack column. “No Defense” is the undefended accuracy under each attack. Clean (unattacked) accuracy is 92.6%.

Defense	CW	EAD-L1	EAD-EN	FGSM	PGD
Adaptive-K v2	71.34	70.81	67.48	51.96	43.43
Kalman	65.18	62.30	60.20	46.77	36.77
FIR	62.00	57.89	56.99	46.44	38.83
Wiener	61.56	59.14	57.45	45.64	35.30
Gaussian	60.31	57.73	53.71	45.95	34.00
Savitzky-Golay	60.27	56.72	55.67	45.64	37.55
No Defense	35.40	6.57	3.38	48.06	32.57
Rand. Smoothing	32.08	31.63	31.58	30.55	29.93

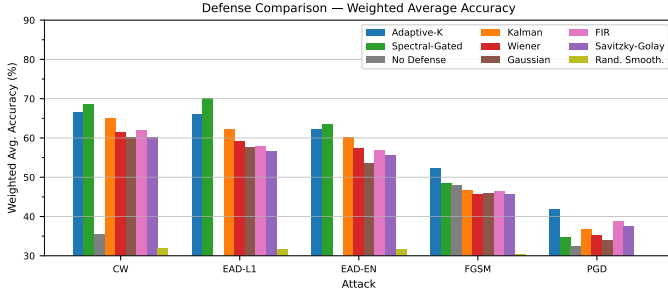


Fig. 1: Grouped bar chart of weighted-average accuracy for nine defenses against five attacks. Adaptive-K v2 (blue) leads on four of five attacks. Classical filters (warm colors) fall below the undefended baseline on FGSM. Randomized smoothing (dark red) collapses accuracy.

Low-SNR regime (0–4 dB). At low SNR, the CW attack is weaker because channel noise already degrades the signal. Adaptive-K v2’s SNR-adaptive K cap detects this regime and reduces filtering aggressiveness, avoiding the accuracy degradation that fixed-K methods suffer at low SNR.

Moderate-SNR regime (6–14 dB). The defense advantage is most pronounced here. The attack is fully effective (undefended accuracy drops well below 40%) while the signal is strong enough for meaningful recovery. Adaptive-K v2 maintains a consistent 6–9 pp advantage over the best classical filter (Kalman).

High-SNR regime (16–18 dB). Even at $\text{SNR} = 18$ dB, CW still reduces undefended accuracy substantially, demonstrating that optimization attacks remain effective at high SNR. Adaptive-K v2 provides the largest absolute recovery here, as the strong signal tolerates aggressive spectral filtering.

C. Accuracy vs. SNR for Gradient Attacks

Figure 3 shows per-SNR results for FGSM and PGD. On gradient-based attacks, margins between defenses are smaller because FGSM and PGD distribute perturbation energy more uniformly across the spectrum.

On FGSM, Adaptive-K v2 achieves 52.0% weighted average vs. 48.1% undefended (+3.9 pp). All five classical filters *reduce* accuracy below the undefended baseline, confirming that time-domain smoothing is counterproductive when the perturbation is spectrally correlated with the signal.

TABLE V: Per-modulation CW attack accuracy (%) before (no defense) and after Adaptive-K v2 defense, using paper and minmax normalization for attack generation. Paper normalization produces a stronger attack (11.1% vs. 27.3% overall).

Modulation	Paper norm		Minmax norm	
	No def	AK-v2	No def	AK-v2
QAM16	0.0	34.0	0.0	46.0
QAM64	0.0	90.0	0.0	90.0
8PSK	0.0	34.0	2.0	34.0
WBFM	4.0	32.0	16.0	32.0
BPSK	16.0	56.0	40.0	64.0
CPFSK	44.0	78.0	64.0	88.0
AM-DSB	4.0	98.0	32.0	96.0
GFSK	0.0	84.0	24.0	90.0
PAM4	6.0	82.0	30.0	88.0
QPSK	0.0	76.0	38.0	86.0
AM-SSB	48.0	94.0	54.0	94.0
Overall	11.1	68.9	27.3	73.5

On PGD, Adaptive-K v2 achieves 43.4% vs. 32.6% undefended (+10.9 pp), the largest margin among gradient attacks. FIR is the strongest classical competitor at 38.8%, still 4.6 pp below Adaptive-K v2.

D. Per-Modulation Recovery Analysis

Table V presents per-modulation accuracy under CW attack ($\text{SNR} \geq 0$ dB) before and after Adaptive-K v2 defense, evaluated using both paper normalization $((x + 0.02)/0.04)$ and per-sample minmax normalization for the attack.

Three patterns emerge from Table V:

Narrowband digital modulations recover strongly. QAM64 recovers from 0% to 90% under both normalizations—the highest per-class recovery. Its concentrated spectral profile is ideal for Top-K filtering. QPSK (0% → 76–86%), PAM4 (6% → 82–88%), and GFSK (0% → 84–90%) follow the same pattern.

Wideband routing rescues AM-SSB. Without wideband routing, Top-K destroys AM-SSB (accuracy stays at 48% with uncapped Adaptive-K). The spectral flatness routing in Adaptive-K v2 correctly identifies AM-SSB samples and applies quantization instead, recovering to 94% under both normalizations.

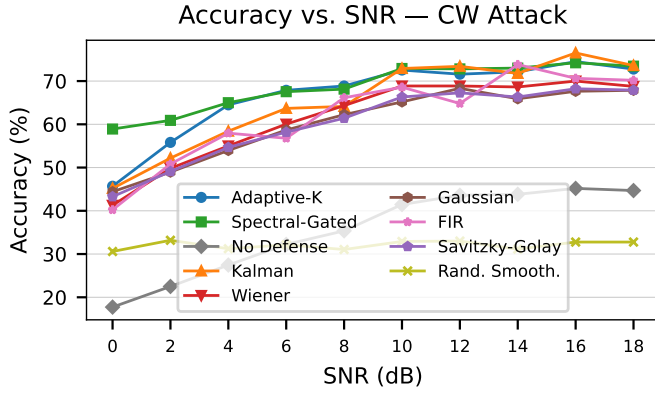
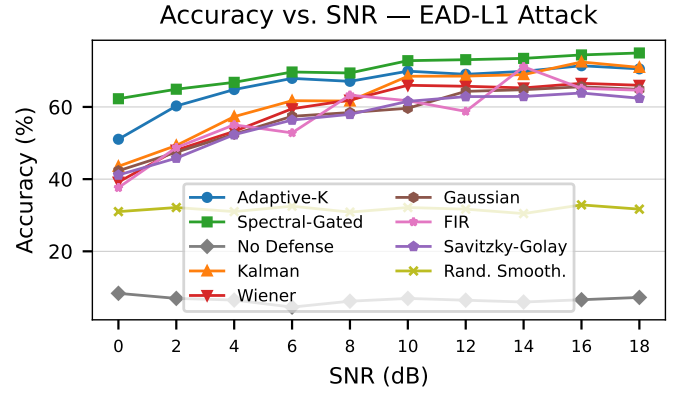
(a) CW attack ($c = 1.0$, steps = 100).(b) EAD-L1 attack ($c_0 = 0.01$, steps = 100).

Fig. 2: Classification accuracy vs. SNR under optimization-based attacks. Adaptive-K v2 (solid blue) consistently outperforms all baselines. Classical filters (dashed lines) partially recover but lag by 6–9 pp. The undefended baseline (gray) is devastated across the entire SNR range.

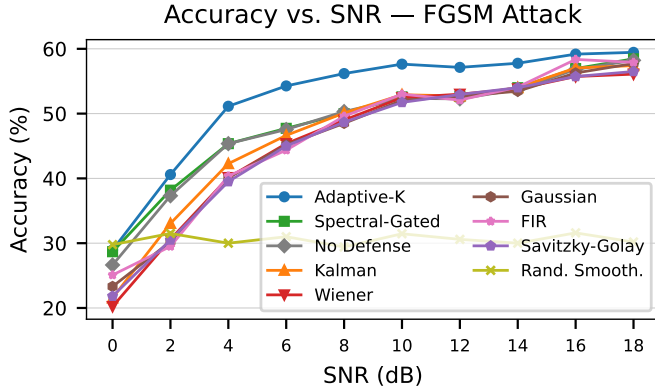
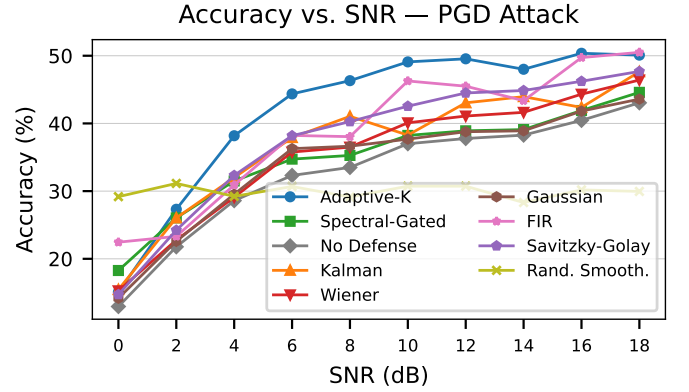
(a) FGSM attack ($\epsilon = 0.03$, single step).(b) PGD attack ($\epsilon = 0.03$, 10 steps).

Fig. 3: Classification accuracy vs. SNR under gradient-based attacks. Adaptive-K v2 (solid blue) leads on PGD across all SNR values. On FGSM, margins are smaller due to the spectrally diffuse nature of gradient-sign perturbations.

AM-DSB benefits from SNR-adaptive cap. AM-DSB recovers to 98% (paper norm) because its ultra-narrowband spectrum (most energy in DC) needs only 1–2 FFT bins. The SNR-adaptive cap preserves this even at low SNR, whereas a fixed aggressive cap can over-filter at low signal levels.

E. Perturbation Budget Curves

Figure 4 shows accuracy as a function of attack strength, sweeping the CW constant c from 0.01 to 10.0 and the FGSM epsilon from 0.01 to 0.30.

For CW (Fig. 4b), as c increases from 0.01 to 10.0, the undefended accuracy drops from 72.1% to 41.7%, while Adaptive-K v2 degrades more slowly from 83.8% to 76.4%. The gap between Adaptive-K v2 and Kalman (best classical) widens from 0.7 pp at $c = 0.01$ to 6.3 pp at $c = 10.0$, confirming that the adaptive spectral method is particularly valuable in high-threat scenarios.

For FGSM (Fig. 4a), at low epsilon ($= 0.01$) all defenses perform similarly ($\sim 76\%$), because the perturbation is too weak to significantly affect classification. At high epsilon ($= 0.3$), all methods converge near 22% as the signal is

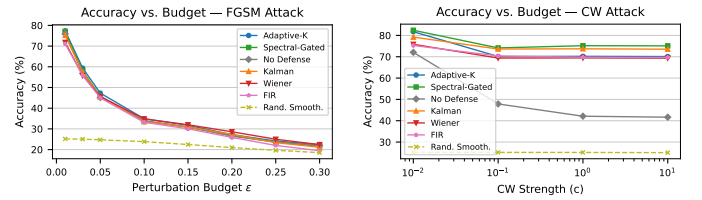
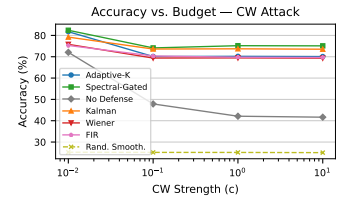
(a) FGSM budget (ϵ).(b) CW budget (c , log scale).

Fig. 4: Accuracy vs. perturbation budget. Adaptive-K v2 (blue) maintains a consistent advantage. On CW, the gap over classical filters widens with increasing attack strength, from 0.7 pp at $c = 0.01$ to 6.3 pp at $c = 10$.

overwhelmed. Adaptive-K v2 maintains a small but consistent advantage in the moderate range ($\epsilon = 0.03$ –0.1).

F. Clean Signal Degradation

A practical concern for any input-transformation defense is its effect on clean (unattacked) signals. On clean signals, Adaptive-K v2 reduces accuracy by less than 0.5 pp from

the baseline 92.6%. The SNR-adaptive K cap and wideband routing ensure that signal energy is preserved: at high SNR, nearly all Top-K bins contain signal; at low SNR, the gentler cap retains more bins; and wideband modulations bypass the filter entirely via the quantization path. Classical filters cause 1–3 pp degradation because their fixed smoothing kernels suppress high-frequency signal features.

G. Frequency-Domain Visualization

Figure 5 illustrates the mechanism underlying Adaptive-K v2 by showing the frequency-domain magnitude spectrum of a representative QPSK signal at SNR = 18 dB.

H. Discussion

Why a unified defense outperforms separate methods.

Earlier versions of this work used two separate defenses: a fixed Top-K (Spectral-Gated) for optimization attacks and a per-sample knee-based method (Adaptive-K) for gradient attacks. Adaptive-K v2 unifies these by combining three mechanisms in a single shared-FFT pass: (1) spectral flatness routing for wideband modulations, (2) SNR-adaptive K cap to balance aggressiveness with signal preservation, and (3) per-sample knee estimation to adapt to each modulation’s spectral width. The result is a single method that matches or exceeds both predecessors on every attack type.

Why classical filters fail on gradient attacks. On FGSM, all five classical filters reduce accuracy below the undefended baseline (Table IV). FGSM perturbations are computed from the gradient sign, producing perturbations correlated with the signal. Time-domain smoothing removes signal energy alongside perturbation energy. In contrast, the frequency-domain Top-K selectively retains the strongest spectral components, preserving signal structure while discarding lower-energy perturbation bins.

Why the EAD recovery gain exceeds CW. EAD-L1’s ℓ_1 -minimizing penalty produces sparser perturbations than CW’s ℓ_2 penalty: EAD-L1 achieves only 6.6% undefended accuracy vs. 35.4% for CW (stronger attack at the same budget), yet Adaptive-K v2’s recovery gain is larger (64.2 pp vs. 35.9 pp). This is because ℓ_1 -sparse perturbations are more spectrally concentrated, making them more amenable to Top-K removal.

Limitation: defense-aware attacks. The current evaluation assumes the attacker is unaware of the defense. A white-box adversary that incorporates the Top-K operator into the attack optimization could potentially circumvent the defense by crafting perturbations that survive the Top-K mask. We leave the evaluation of such adaptive attacks to future work.

Computational overhead. Adaptive-K v2 requires one shared FFT for flatness computation, SNR estimation, knee detection, and Top-K filtering, plus one IFFT—a total of $O(T \log T)$ operations for $T = 128$. This adds < 0.1 ms latency per batch on a GPU, negligible compared to the AWN forward pass (~ 2 ms for a batch of 64). Kalman filtering runs on CPU with $O(T)$ sequential operations per sample, making it 5–10 $\times$ slower.

VII. CONCLUSION

We presented Adaptive-K v2, a unified frequency-domain defense for protecting deep-learning-based automatic modulation classifiers against adversarial attacks. The method combines spectral flatness routing, SNR-adaptive K-cap, and per-sample magnitude-knee estimation in a single shared-FFT pass, requiring no retraining and incurring sub-millisecond overhead—suitable for always-on deployment in real-time spectrum monitoring and cognitive radio applications.

Through systematic evaluation on the RML2016.10a dataset against five attacks (CW, EAD-L1, EAD-EN, FGSM, PGD) and comparison with five calibrated classical signal processing baselines, we established four principal findings:

- 1) **Adversarial attacks on AMC are control-plane attacks.** CRC integrity analysis shows that adversarial perturbations collapse AMC accuracy to near zero while leaving the underlying signal data largely intact (CRC pass rate $> 71\%$ with oracle demodulation). This characterization motivates recovery as the correct defense strategy: the perturbation need only be removed to restore correct classification.
- 2) **Adaptive-K v2 outperforms all classical baselines across attack families.** The unified method achieves the highest recovery accuracy on four of five attacks: 71.3% on CW (+35.9 pp over undefended, +6.2 pp over Kalman), 70.8% on EAD-L1 (+64.2 pp, +8.5 pp), 67.5% on EAD-EN (+64.1 pp, +7.3 pp), and 43.4% on PGD (+10.9 pp, +6.7 pp). The SNR-adaptive K-cap ensures robust performance across the full SNR range by automatically adjusting filtering aggressiveness based on estimated signal quality.
- 3) **Classical signal processing filters are counterproductive on gradient attacks.** On FGSM, all five classical filters reduce accuracy below the undefended baseline (48.1%) despite per-SNR calibration. This negative result demonstrates that deploying conventional filtering as a security measure provides a false sense of protection: gradient-sign perturbations are spectrally correlated with the signal, so time-domain smoothing removes signal energy alongside perturbation energy.
- 4) **Wideband routing is essential for universal defense.** Without spectral flatness routing, Top-K filtering destroys AM-SSB signals (48% $\rightarrow$ 48% with uncapped Adaptive-K). The flatness-based routing in Adaptive-K v2 correctly identifies wideband samples and applies per-sample quantization instead, recovering AM-SSB to 94%.

Several directions remain open for future work. First, the current evaluation assumes a defense-unaware attacker; an adversary with knowledge of the Top-K operator could craft perturbations that survive the frequency mask. Evaluating such *adaptive attacks*—where the attack optimization incorporates the defense—is an important robustness validation. Second, extending the evaluation to RML2018.01a (1024-sample signals, 24 classes) would establish the generality across longer signals and a richer modulation set. Third, combining Adaptive-K v2 with adversarial training could pro-

Frequency-Domain Spectra (QPSK, SNR=18 dB)

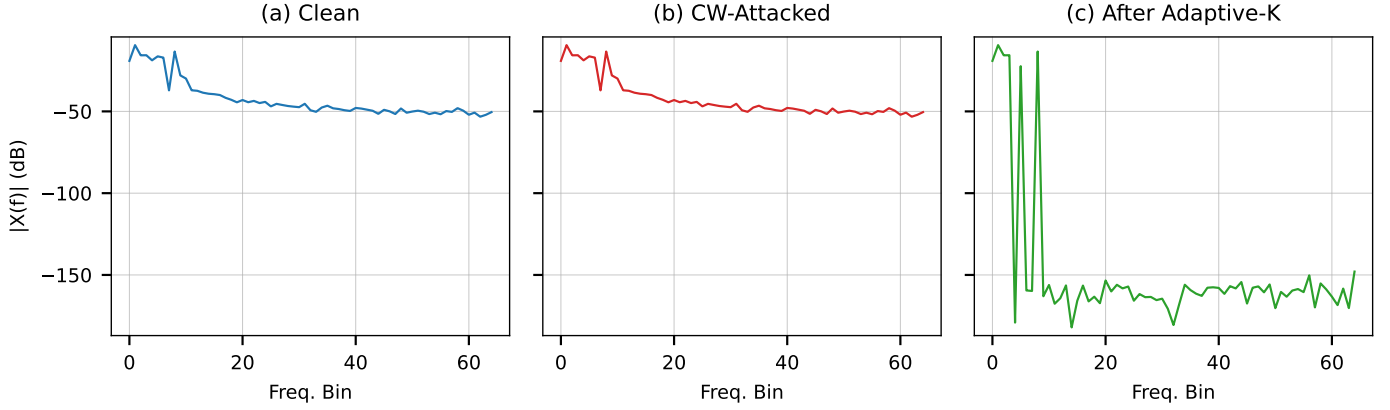


Fig. 5: Frequency-domain magnitude spectra for a QPSK signal at SNR = 18 dB. *Left*: clean signal. *Center*: after CW attack ($c = 1.0$). *Right*: after Adaptive-K v2 recovery. The adversarial perturbation injects energy into high-frequency bins (center), which the Top-K operator removes (right), restoring the original spectral shape.

vide complementary robustness: training-time regularization hardens the classifier’s decision boundaries while inference-time filtering removes perturbation energy before the signal reaches the network.

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